

of a sequence of actions that can be different for different problems.

- d. Identify the problem category.
- e. Find the appropriate resolution.
- f. Identify the action sequence to resolve the problem.
- g. Record the action sequence for resolution.

While steps (a) to (c) are ideal for traditional RPA since they typically deal with structured data entered through forms, steps (d) to (g) require dealing with unstructured textual content, understanding contextual knowledge, as well as content analytics and judgement. NLP modules aid in problem categorisation based on unstructured text, knowledge mining of back-office problem manuals to identify the resolution, parsing previous interactions to mine the action sequence in terms of extracting the verbs/ subject/object from the text, and outputting the necessary action sequence. Existing RPA solutions can help automate steps (a) to (c), but typically fail when automating steps (d) to (e). However, by employing natural language processing techniques, RPA tool suites can be enhanced to automate steps (d) to (e).

A similar problem emerges in insurance claim validation. Typically, insurance claims contain unstructured data like supporting documents that need to be analysed in case of any potential violations. Data also needs to be extracted from multiple systems and validated against each other. Given

that it is possible to synthesise a set of rules to cover the validation process, this is a good candidate for RPA.

So far, we have been talking about existing RPA capabilities in terms of being able to handle static rule-based tasks which are repetitive and whose decision-making can be condensed into a form of *if-then* statements. How can RPA deal with more complex decision-making scenarios? How can it learn to handle exceptions in a workflow like humans do? Who are the leading vendors in RPA? We will cover these topics in next month's column.

If you have any favourite programming questions/ software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, here's wishing all our readers a wonderful and productive new year ahead! 

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